in one file, some in another. If nothing else you will need to flip between two editor windows.

Consider for a moment a Web page as an object—that is, in an object-oriented way. On the surface, a Web page is a pair of HTML tags containing HEAD tags and BODY tags. Regardless of the design or content of the page, these tags must exist, and inside them will be placed further tags. Inside the BODY tags a table can be placed for controlling the layout of the page. Inside the cells of the table are either links to other pages on the site or some content unique to the page.

FreeEnergy is a system that attempts to encapsulate major pieces of each page into files to be included on demand. Before we proceed, we want to state motivations clearly. Our first concern when developing a Web site is that it be correct and of the highest quality. Second is that it may be developed and maintained in minimal time. After these needs are addressed, we consider performance. Performance is considered last because of the relatively cheap cost of faster hardware. Moore's law suggests that eighteen months from now, CPU speed and memory capacity will have doubled for the same price. This doubling costs nothing but time. Also, experience has shown that a small minority of code contributes to a majority of the time spent processing.

The FreeEnergy system uses more calls to include than you'd find where you are simply making a few includes at the top of your pages. Hits to the file system do take longer than function calls, of course. You could place everything you might need in one large file and include it on every page, but you will face digging through that large file when you need to change anything. A trade has been made between the performance of the application and the time it takes to develop and maintain it.

We called this system FreeEnergy because it seems to draw power from the environment that PHP provides. The include function in PHP is quite unique and central to FreeEnergy, especially the allowance for naming a script with a variable. The content unique to a page is called a screen. The screen name is passed to a single PHP script, which references the screen name in a large array that matches the screen to corresponding layout and navigation modules.

The FreeEnergy system breaks Web pages into five modules: action, layout, navigation, screen, and utility. Action modules perform some sort of write function to a database, a file, or possibly to the network. Only one action module executes during a request, and they are executed before the screen module. An action module may override the screen module named in the request. This is helpful in cases where an action module is attempting to process a form and the submitted data are incomplete or otherwise unsatisfactory. Action modules never send data directly to the screen. Instead, they add messages to a stack to be popped later by the layout module. It is possible that an action module will send header information, so it's important that no output be produced.